

Week 1 Flush+Reload Calibration Report

Deliverable summary

This calibration establishes a timing threshold that separates a cache **hit** from a cache **miss** for the Week 1 Flush+Reload probe.

Probe variant	Hit median	Miss median	Selected threshold	Classification rule	Observed error
RDTSC control	78 ticks	1068 ticks	150 ticks	<code>< 150 = hit; >= 150 = miss</code>	1 false miss, 0 false hits; 0.001% overall
RDTSCP formal	100 ticks	1094 ticks	170 ticks	<code>< 170 = hit; >= 170 = miss</code>	0 false misses, 0 false hits; 0% overall

The **formal Week 1 threshold is 170 TSC ticks with RDTSCP + LFENCE**. The RDTSC result is retained as a control experiment to quantify timestamp-instruction overhead.

Probe and measurement procedure

The probe uses a 64-byte cache line in a shared, read-only, file-backed `MAP_SHARED` mapping. Each sample is prepared independently:

```
1 flush target with clflush
2 -> mfence
3 -> flush-settle busy-wait (5000 NOPs)
4 -> HIT condition: one untimed load
5     MISS condition: no access to the target
6 -> reload-settle busy-wait (5000 NOPs)
7 -> timed reload
8 -> store the result in memory
```

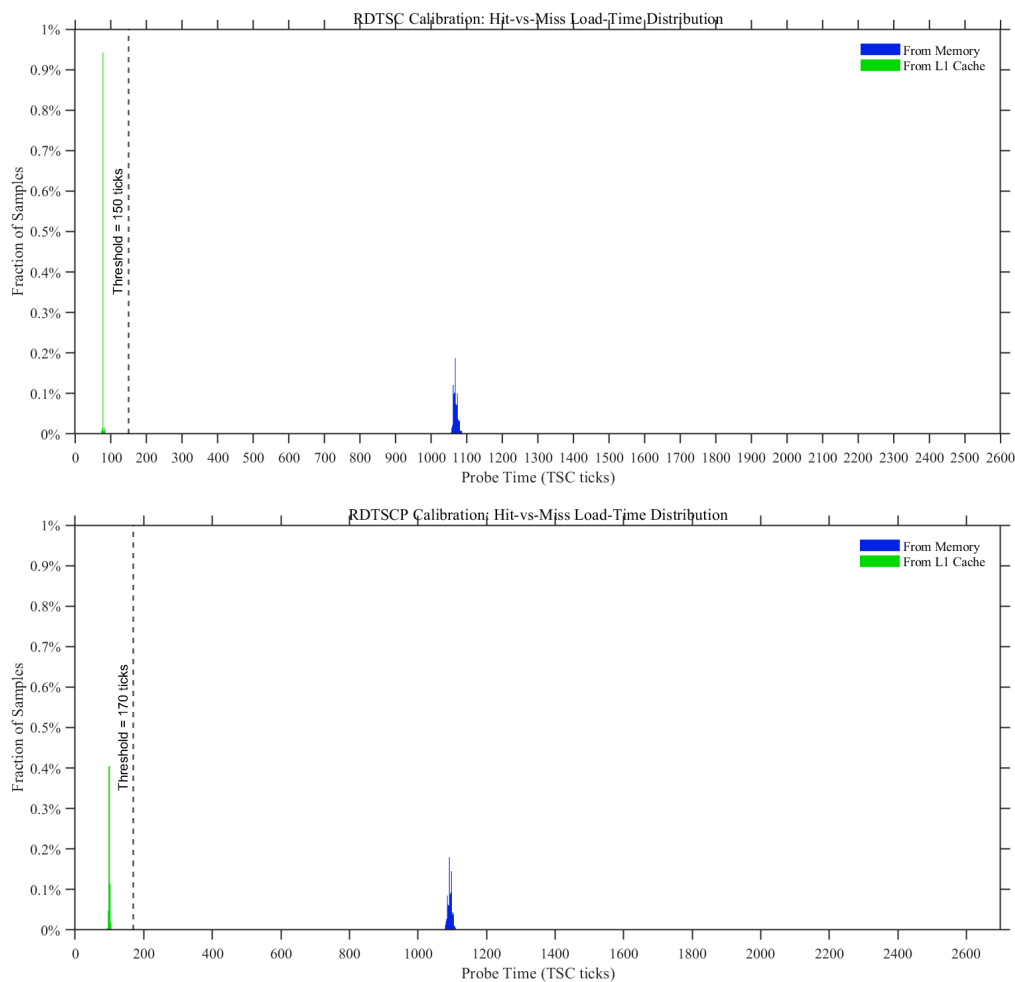
The timed reload is fenced so that the timestamp interval encloses the target load. The formal timer is:

```
1 lfence
2 rdtscp
3 lfence
4 load target
5 lfence
6 rdtscp
7 lfence
8 subtract timestamps
```

The RDTSC control uses the equivalent LFENCE-delimited sequence with `RDTSC` in place of `RDTSCP`. In both cases, samples are buffered and written after measurement rather than printing or performing file I/O inside the timing loop.

Calibration histograms

RDTSC control experiment



The green hit distribution is concentrated near 98–100 ticks; 92.32% of hit samples lie between 98 and 102 ticks. The blue miss distribution is centered near 1092–1094 ticks, with a minimum of 824 ticks. The selected threshold of **170 ticks** lies in the wide empty gap between the two distributions: it is above the largest observed hit (156 ticks) and below the smallest observed miss (824 ticks). Consequently, the calibration contains zero observed false misses and zero observed false hits.

Why these thresholds were chosen

The threshold was chosen from the empirical separation of the two measured distributions, not from an assumed cache-latency constant.

1. It is placed above the complete observed hit cluster, including high-latency hit samples.
2. It remains far below the minimum observed miss, leaving a large safety margin.
3. It is defined separately for each timestamp instruction because `RDTSCP` has additional measurement overhead (`TSC_AUX/ECX` read and stronger ordering behavior). The observed hit-median difference is 22 ticks (78 for `RDTSC` versus 100 for `RDTSCP`).
4. The `RDTSCP` threshold is used for the formal Week 1 result because it provides a stronger timestamp-ordering boundary and can additionally expose CPU migration through `TSC_AUX`.

The threshold is therefore a classifier for this calibrated probe and environment; it is not interpreted as a universal boundary for every CPU, timer sequence, or runtime configuration.

Noise sources considered and controls applied

CPU placement and shared-core interference

- The experiment ran on Intel Core Ultra 7 155H Meteor Lake, using one Redwood Cove P-core.
- The process was explicitly pinned to **CPU 5** with `taskset -c 5 / sched_setaffinity`.
- CPUs 1–5 were isolated with `isolcpus=1-5`; ordinary housekeeping work used CPUs 0, 6, and 7.
- SMT/Hyper-Threading was disabled, removing sibling-thread contention for private caches, queues, and execution resources.

Scheduling, interrupts, and idle-state variation

- `nohz_full=1-5` reduced periodic scheduler ticks on isolated CPUs.
- `rcu_nocbs=1-5` moved RCU callbacks away from the measurement cores.
- Ethernet IRQs were redirected to housekeeping CPUs (0, 6, and 7); Wi-Fi was kept down and `irqbalance` was inactive.
- `processor.max_cstate=1`, `intel_idle.max_cstate=1`, and `idle=poll` limited deep idle/wake-up variation.
- The `performance` governor was selected and Turbo Boost was disabled (`intel_pstate/no_turbo=1`) to reduce frequency transitions.

Cache-state and microarchitectural contamination

- `clflush` followed by `mfence` was used to create the miss state.
- The probe no longer performs an implicit flush at its end. Cache-state preparation is explicit at the beginning of each sample, avoiding consecutive flushes and cross-sample coupling.
- The untimed HIT load and the timed reload were separated by **5000 NOPs** so that line fill and cache allocation could settle before measurement.
- A **20,000-NOP inter-round gap** separated the HIT and MISS samples of adjacent rounds, reducing residual effects from flushes, fills, fences, and load queues.
- No `sleep`, `usleep`, `sched_yield`, system calls, terminal output, or file writes occurred in the timing loop.
- The mapped page, result buffers, and timing path were warmed up before collecting samples; allocations and page faults were avoided during the loop.
- Hardware-prefetcher state (`MSR 0x1A4`) was recorded rather than silently changed for the basic Week 1 calibration.

Timestamp and data-processing controls

- The same fenced timestamp structure was used for HIT and MISS.
- The TSC is invariant (`constant_tsc / nonstop_tsc`); all reported values are TSC ticks, not instantaneous core cycles.
- `RDTSCP` was paired with `LFENCE` for the formal measurement. Its `TSC_AUX` value can be checked to detect migration.
- 50,000 HIT and 50,000 MISS samples were collected for each timer variant.

- Histograms used 2-tick bins and probability normalization. Error rates were computed as:

```
1 | false-miss rate = fraction of HIT samples >= threshold
2 | false-hit rate  = fraction of MISS samples < threshold
```

Result and limitation

The calibration demonstrates a stable, clearly separated Flush+Reload timing primitive. The formal RDTSCP configuration with a **170-tick threshold** achieved **0/100,000 observed classification errors** on this dataset. Remaining asynchronous events (for example NMI, SMI, IPI, unavoidable hardware IRQs, page faults, or occasional kernel activity) may still appear as outliers, so the threshold should be re-calibrated if the CPU, timer sequence, affinity, frequency policy, or kernel configuration changes.